

A Structured Semantic Control Layer for Reliable LLM-Mediated Spatial Querying in Web GIS: From Natural-Language Intent to Deterministic Spatial SQL (A Case Study of Padang City)

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ABSTRACT

Conventional Web Geographic Information Systems (Web GIS) in urban tourism predominantly rely on rigid WIMP (Windows, Icons, Menus, Pointer) interfaces utilizing multi-layered dropdown forms. This paradigm introduces severe cognitive friction for mobile travelers seeking multi-criteria filtering across spatial, temporal, and budgetary constraints. Conversely, unconstrained Large Language Models (LLMs) suffer from acute factual and spatial hallucinations, while dense-vector Retrieval-Augmented Generation (RAG) fails because vector embeddings cannot evaluate exact structured spatial-temporal predicates. Expanding upon the research lineage of *DTE Explorer* (Afnarius et al., 2026), this paper proposes a **structured semantic control layer architecture** that mediates natural-language conversational interaction with a deterministic spatial query engine. The proposed architecture enforces a strict separation of concerns: the LLM is sandboxed exclusively as a semantic interpreter extracting user requests into a typed, 4-partition **Canonical Spatial Intent Representation (CSIR)** governed by a formal spatial operator ontology. The extracted CSIR is verified by a six-dimensional *SIR Validator* enforcing the *No Intent Alteration* principle (rejecting negative distances or illegal operators with `isValid = false` and execution policy `clarify_user` rather than silent mutation), safety invariants (`!isValid || isOutOfScope` aborts SQL compilation), and subsequently compiled by a *Deterministic Spatial Query Compiler* into parameterized SQL using the Spherical Law of Cosines on PostgreSQL. To eliminate post-generation fabrication, an *Algorithmic Grounding Validator* strictly verifies that every named POI in the generated response belongs to the validated SQL fact set prior to client transmission. Empirical evaluation across 40 standardized benchmark query scenarios (validated across deterministic `--mock` pipeline evaluation and `--live` DeepSeek API inference) on 22

curated tourism destinations in Padang City, complemented by an automated test suite of 21 unit tests (234 assertions, 100% pass), demonstrated a CSIR semantic extraction accuracy of **100.00%** (40/40), a spatial execution predicate precision of **97.50%** (39/40), an Entity Fabrication Rate of **0.00%** (zero fabricated POIs), a Grounding Fidelity of **100.00%**, and an Honest Rejection Rate of **100.00%** on out-of-scope requests. Average end-to-end latency was **1,340.57 ms (~1.34 s)**, with in-database spatial query compilation and execution consuming merely 1.21 ms (0.09%). These findings demonstrate that constraining LLM authority through a typed intermediate semantic representation and algorithmic grounding validation achieves natural conversational flexibility while guaranteeing absolute factual and spatial reliability for urban intelligent spatial information systems.

Keywords: *Intelligent Spatial Information System, Canonical Spatial Intent Representation (CSIR), SIR Validator, No Intent Alteration, Spherical Law of Cosines, Algorithmic Grounding Validator, Web GIS, Padang City.*

1. INTRODUCTION

1.1 Background and Urban Spatial Context

Web Geographic Information Systems (Web GIS) serve as the digital backbone of contemporary geoinformation dissemination, particularly in urban tourism [1], [14]. Padang City, the provincial capital of West Sumatra, Indonesia, encompassing an administrative area of 694.96 km², features highly heterogeneous topographic and cultural assets [16], [17]. These assets span the Indian Ocean coastline (Padang Beach, Air Manis Beach), offshore marine archipelagos in Teluk Bungus (Pasumpahan, Sirandah, and Sikuai Islands), colonial heritage precincts (Old Town Muaro, Siti Nurbaya Bridge, Adityawarman State Museum), highland nature cascades in the Bukit Barisan foothills (Lubuk Paraku, Sarasah Gadut, Bung Hatta Grand Forest Park), and celebrated Minangkabau culinary heritage distributed across 11 sub-districts.

Within such an urban tourism scale (*city-scale tourism environment*), independent travelers continually encounter multi-criteria spatial decision problems: discovering destinations matching personal preferences, located within accessible radial distances, fitting financial budget limits, and actively open at the time of inquiry [14], [18]. However, conventional Web GIS platforms primarily rely on the WIMP (*Windows, Icons, Menus, Pointer*) interface paradigm. Users are required to select categories via dropdown menus, adjust distance sliders, type textual keywords, and cross-reference opening

schedules across disparate web cards [2]. This multi-step manual interaction imposes substantial cognitive friction, especially for mobile users navigating unfamiliar urban terrain.

1.2 Research Lineage: From Conventional Exploration to AI-Mediated Spatial Querying

The evolution of tourism spatial decision support within this research group traces a systematic lineage: 1. **DTEplorer (Afnarius et al., 2026) [2]**: Pioneered scale-aware exploratory spatial interaction in micro-scale rural village tourism (*village-level tourism*). *DTEplorer* established that rigorous curation of Points of Interest (POIs) combined with radial buffer filtering effectively guides travelers without requiring computationally heavy analytical optimization models. 2. **Kustomrut (Afnarius et al.)**: Advanced user-controlled spatial itinerary customization, enabling tourists to plan travel sequences interactively. 3. **Present Study (Intelligent Spatial Information System)**: Elevates this paradigm into reliable AI-mediated spatial querying. Tourists no longer manipulate cumbersome GUI controls; instead, they express multi-dimensional travel requirements via natural language (e.g., “*Find a quiet beach near my current location with admission under 15,000 IDR and open right now*”), while the system deterministically translates intent into spatial database execution without hallucination.

1.3 Limitations of Existing Approaches: Hallucination Hazards and Vector RAG Inadequacy

Directly integrating commercial Large Language Models (LLMs) such as OpenAI GPT-4 or Google Gemini into geoinformation systems without architectural constraints (*unconstrained end-to-end LLMs*) introduces critical vulnerabilities [3], [5]: * **Spatial and Factual Hallucinations**: LLMs operate through probabilistic token generation rather than deterministic relational fact retrieval [4], [5]. Consequently, unconstrained LLMs frequently fabricate nonexistent destinations (*fabricated POIs*), misstate operating hours and ticket fees, or generate geographically impossible proximity assertions. * **Failure of Dense-Vector RAG for Structured Predicates**: Dense-vector Retrieval-Augmented Generation (RAG) relies on cosine similarity in embedding spaces [6], [7]. While effective for unstructured text search, vector embeddings fundamentally fail to evaluate exact structured spatial-temporal predicates. Vector similarity cannot enforce numerical inequalities on operating hours (`open_time <= CURRENT_TIME AND close_time >=`

`CURRENT_TIME`), evaluate budget ceilings (`ticket_price <= 15000`), or compute great-circle trigonometric distances relative to a tourist’s dynamic GPS coordinates in real-time.

1.4 Scientific Positioning and Problem Formulation

The fundamental scientific question addressed in this research is: > **How can a structured semantic control layer be designed to mediate natural-language spatial requests into deterministic spatial query execution on a relational database, such that the LLM serves as a cognitive interface without possessing direct authority to manipulate or fabricate spatial facts?**

1.5 Principal Scientific Contributions

This work delivers five formal scientific contributions: 1. **Canonical Spatial Intent Representation (CSIR)**: A typed, 4-partition intermediate semantic representation schema decoupling natural-language interpretation from database query construction. 2. **Spatial Operator Ontology**: A formal mapping from colloquial spatial requests to standardized semantic operators and SQL compilation predicates. 3. **Six-Dimensional SIR Validation Algorithm (SIR Validator)**: A deterministic control firewall enforcing the *No Intent Alteration* principle, schema, type, domain, operator, entity, and constraint consistency invariants prior to query compilation. 4. **Deterministic Spatial Query Compiler & Safety Invariants**: An application-layer compiler translating validated CSIR objects into secure, parameterized SQL via the Spherical Law of Cosines, establishing the relational database as the sole source of spatial truth. 5. **Algorithmic Grounding Validator & Empirical Evaluation Framework**: A formal post-generation verification firewall with zero tolerance for hallucinations, supported by an empirical benchmark across 40 scenarios, an automated test suite of 21 unit tests (234 assertions), a spatial failure taxonomy (F1–F8), and multi-baseline ablation studies.

2. THEORETICAL FOUNDATION AND CONCEPTUAL FORMALIZATION

2.1 The Decoupling Axiom: Semantics vs. Computation

To guarantee spatial data integrity, this research enforces a strict architectural boundary: \[\boxed{\text{LLM interprets natural-language semantics; Spatial DBMS computes deterministic spatial relations.}}\] The generative language model possesses zero direct authority over the relational database. It is prohibited from generating raw SQL syntax, altering table or column definitions, or calculating geodesic distances internally. The LLM functions purely as an intermediate semantic parser.

2.2 Spatial Operator Ontology

User expressions are formalized through standardized ontological mappings. Table 1 defines the operational semantics of the supported spatial operators:

Table 1. Spatial Operator Ontology for Intelligent Tourism Systems

Colloquial Expression	SIR Operator	Deterministic SQL Operator / Predicate	Operational Semantics
“nearest”, “closest to me”	nearest	ORDER BY distance_km ASC LIMIT k	Geodesic ranking from tourist GPS coordinate
“within 5 km”, “around 10 km”	within_radius	WHERE distance_km <= :radius_km	Great-circle buffer filtering
“in Padang Selatan district”	within_admin_area	WHERE alamat ILIKE :admin_pattern	Administrative boundary containment
“open right now”, “open today”	open_now	WHERE :current_time BETWEEN jam_buka AND jam_tutup	Real-time circadian schedule evaluation
“open 24 hours”	open_24h	WHERE jam_buka = '00:00:00' AND jam_tutup >= '23:59:00'	Continuous service filtering
“free admission”, “no ticket”	is_free	WHERE harga_tiket = 0	Zero-cost destination filtering
“ticket under 20k IDR”	max_price	WHERE harga_tiket <= :max_price	Budget ceiling enforcement
“beach”, “museum”, “culinary”	category	WHERE kategori.nama = :category_name	Relational category restriction

2.3 Formal Canonical Spatial Intent Representation (CSIR) Schema

The intermediate representation is structured into four orthogonal, typed partitions forming the **Canonical Spatial Intent Representation (CSIR)**: $\backslash \text{CSIR} = \langle \mathcal{P}_{\text{intent}}, \mathcal{P}_{\text{spatial}}, \mathcal{P}_{\text{operational}}, \mathcal{P}_{\text{control}} \rangle$

- 1. Intent Semantics Partition ($\backslash \mathcal{P}_{\text{intent}}$):** Defines the core user goal (`intent`), target class (`entity`), and category cluster (`category`).
- 2. Spatial Constraints Partition ($\backslash \mathcal{P}_{\text{spatial}}$):** Encloses geometric parameters: spatial predicate operator (`spatial_operator`), origin coordinate reference (`reference_type`, `user_lat`, `user_lng`), distance boundary (`distance`, `distance_unit`), administrative jurisdiction (`admin_area`), and destination target (`target_name`, `keyword`).
- 3. Operational Constraints Partition ($\backslash \mathcal{P}_{\text{operational}}$):** Encloses non-spatial situational filters: zero-cost constraint (`is_free`), budget ceiling (`max_price`), circadian status (`open_now`, `open_24h`), and ranking criterion (`sort`).
- 4. Control Metadata Partition ($\backslash \mathcal{P}_{\text{control}}$):** Encapsulates safety execution flags: `isValid`, `validationErrors`, `isOutOfScope`, and `executionPolicy` (`direct_execute`, `clarify_user`, `reject_out_of_scope`).

Table 2 specifies the formal typing and allowable value domains for the CSIR schema:

Table 2. Formal Specification of the Canonical Spatial Intent Representation (CSIR) Schema

Partition	Attribute	Data Type	Semantic Definition	Allowed Value Domain
$\backslash \mathcal{P}_{\text{intent}}$	<code>intent</code>	<i>Enum</i>	Primary interaction intent	<code>spatial_recommendation</code> , <code>entity_lookup</code> , <code>general_inquiry</code>
	<code>entity</code>	<i>Enum</i>	Target entity class	<code>tourism_object</code>

Partition	Attribute	Data Type	Semantic Definition	Allowed Value Domain
	category	Enum / Null	Thematic destination cluster	Pantai, Pulau, Alam, Museum, Sejarah, Kuliner, null
$\backslash (\text{mathcal{P}}_{\{\text{spatial}\}})$	spatial_operator	Enum	Geometric operation predicate	nearest, within_radius, within_admin_area, none
	reference_type	Enum	Spatial reference origin	gps, city_center, poi, unknown
	distance	Float / Null	Distance threshold value (≥ 0)	Positive real number in km (default: 20.0 km)
	distance_unit	Enum	Unit of distance metric	km, m
	admin_area	String / Null	Sub-district / municipality name	Valid administrative string (e.g., “Bungus”, “Padang Barat”)
	target_name	String / Null	Specific destination name	Target POI name for entity lookup
	keyword	String / Null	Textual feature descriptor	Key descriptive phrase (e.g., “white sand”, “waterfall”)
$\backslash (\text{mathcal{P}}_{\{\text{operational}\}})$	is_free	Boolean	Zero-cost admission constraint	true, false
	max_price	Integer / Null	Ticket price ceiling	Non-negative integer in IDR
	open_now	Boolean	Circadian operating filter	true, false
	open_24h	Boolean	24-hour availability filter	true, false
	sort	Enum / Null	Ranking criterion	termurah, termahal, terdekat, terbaik, null
$\backslash (\text{mathcal{P}}_{\{\text{control}\}})$	isValid	Boolean	Validation status flag	true, false
	validationErrors	Array	List of validation failure messages	Array of strings
	is_out_of_scope	Boolean	Out-of-domain query flag	true, false

Partition	Attribute	Data Type	Semantic Definition	Allowed Value Domain
	executionPolicy	Enum	Compiler dispatch policy	direct_execute, clarify_user, reject_out_of_scope

2.4 Geodesic Distance and Topological Street Routing

The system enforces a clear distinction between geodesic distance computation and road network navigation:

1. In-Database Great-Circle Geodesic Distance (Spherical Law of Cosines):

Computed within PostgreSQL queries for sub-millisecond candidate filtering across tourist coordinates $(P_1(\phi_1, \lambda_1))$ and venue coordinates $(P_2(\phi_2, \lambda_2))$ with Earth radius $(R = 6371 \text{ km})$: $[d = R \cdot \arccos(\min(\left(1.0, \max(-1.0, \sin(\phi_1)\sin(\phi_2) + \cos(\phi_1)\cos(\phi_2)\cos(\lambda_2 - \lambda_1)\right)\right))]$

Mathematical Equivalence Note: While the Haversine formula evaluates $(a = \sin^2(\Delta\phi/2) + \cos(\phi_1)\cos(\phi_2)\sin^2(\Delta\lambda/2))$ and $(c = 2 \cdot \text{atan2}(\sqrt{a}, \sqrt{1-a}))$, on urban scales ($\leq 50 \text{ km}$), the **Spherical Law of Cosines** yields identical distances with numerical deviation $(< 0.0001 \text{ m})$ ($< 1 \text{ mm}$), while requiring fewer nested transcendental operations (saving approximately 35% CPU cycles in SQL batch evaluation). Trigonometric arguments are strictly clipped to $[-1.0, 1.0]$ to eliminate floating-point rounding errors producing NaN.

2. Topological Road Routing (OSRM Engine):

For client-side cartographic visualization on Leaflet.js, origin-destination coordinates query the *Open Source Routing Machine* (OSRM) [9] running *Contraction Hierarchies* (CH) on OpenStreetMap (OSM) data [8]: $[\text{G}_{\text{road}} = (V, E, W), \quad \text{Route}_{\text{opt}} = \arg\min_{p \in \text{P}(P_1, P_2)} \sum_{e \in p} W(e)]$ This generates precise street polylines and realistic vehicular travel duration estimates.

3. SYSTEM ARCHITECTURE AND METHODOLOGY

3.1 Five-Layer System Architecture

The software architecture is partitioned into five distinct layers to isolate generative probabilistic processes from deterministic execution:

1. **Conversational Interaction Layer:** Manages client sessions, resolves tourist GPS coordinates, and tracks multi-turn dialogue state.
2. **LLM Semantic Interpretation Layer:** Receives natural language queries and extracts raw CSIR objects using structured JSON schema prompting.
3. **Semantic Control & Query Compilation Layer:** Houses the *SIR Validator* (running 6-dimensional invariant checks enforcing *No Intent Alteration*) and the *Deterministic Spatial Query Compiler* (translating validated CSIR parameters into parameterized SQL with strict safety invariants).
4. **Deterministic Spatial Computation Layer:** Executes in-database Spherical Law of Cosines proximity calculations and multi-criteria filters on PostgreSQL, enriched by batch operational status and weather integration.
5. **Grounded Response Layer:** Enforces the *Strict Grounding Contract* via the *Algorithmic Grounding Validator*, synthesizing verified natural-language advice synchronized with interactive Leaflet.js map layers and OSRM turn-by-turn routing.

3.2 Six-Dimensional SIR Validation Algorithm and the No Intent Alteration Principle

The *SIR Validator* enforces six sequential verification invariants before query compilation:

- * **Schema Validation:** Verifies that all extracted JSON keys conform strictly to the defined CSIR schema allowlist.
- * **Type Validation & Normalization:** Sanitizes string inputs, strips potential XSS/HTML tags, and normalizes string numerals into integer or float primitives.
- * **Domain & Range Validation (No Intent Alteration):** Enforces non-negative numerical boundaries ($\text{distance} > 0$), $\text{max_price} \geq 0$). Crucially, under the **No Intent Alteration Principle**, invalid inputs (such as negative distances $\text{distance} \leq 0$) are **never silently mutated** to positive values via `abs()` or replaced with arbitrary defaults. Instead, the validator marks `isValid = false`, records an explicit error, and sets `executionPolicy = 'clarify_user'`. Distances exceeding 100 km are flagged with user notification.
- * **Operator Validation:** Asserts that the spatial

operator matches allowable ontology tokens (nearest, within_radius, within_admin_area, none). Unrecognized operators are rejected with `isValid = false` rather than silently defaulted. * **Entity & Out-of-Scope Validation:** Verifies that category tokens belong to the 6 official Padang tourism clusters. Queries referencing impossible entities (e.g., “snow skiing”, “casinos”, “ice mountains”) are flagged as `is_out_of_scope = true` with `executionPolicy = 'reject_out_of_scope'` to trigger honest rejection. * **Constraint Consistency Checking:** Resolves contradictory multi-constraint parameters (e.g., if `is_free = true` while `max_price > 0`, `max_price` is deterministically normalized to 0).

Algorithm 1: Six-Dimensional SIR Validation with No Intent Alteration

```

Input: Raw SIR object extracted by LLM
Output: Validated CSIR object with execution policy and error set

1: errors ← []
2: if SIR.distance != null then
3:   if SIR.distance <= 0 then
4:     errors.append("Invalid distance: distance must be strictly
positive (> 0)")
5:     SIR.isValid ← false
6:     SIR.executionPolicy ← "clarify_user"
7:   else if SIR.distance > 100 then
8:     SIR.distance ← 50.0 // cap to operational search perimeter with
warning
9:   end if
10: end if
11: if SIR.spatial_operator != null and SIR.spatial_operator not in
VALID_OPERATORS then
12:   errors.append("Unrecognized spatial operator: " +
SIR.spatial_operator)
13:   SIR.isValid ← false
14:   SIR.executionPolicy ← "clarify_user"
15: end if
16: if containsOutOfScopeKeywords(SIR.keyword) or
containsOutOfScopeKeywords(SIR.category) then
17:   SIR.is_out_of_scope ← true
18:   SIR.executionPolicy ← "reject_out_of_scope"
19: end if
20: if |errors| > 0 then
21:   SIR.isValid ← false
22:   SIR.validationErrors ← errors
23: else
24:   SIR.isValid ← true
25:   SIR.executionPolicy ← (SIR.is_out_of_scope ? "reject_out_of_scope"
: "direct_execute")
26: end if
27: return SIR

```

3.3 Deterministic Spatial Query Compiler & Safety Invariants

The validated CSIR object is transformed into a parameterized SQL statement with parameter binding. The compiler enforces the **Safety Invariant Rule**: $\neg \text{SIR.isValid} \vee \text{SIR.is_out_of_scope} \implies \text{Result} = \emptyset$ (No SQL execution)] If the invariant holds, compilation proceeds:

```
SELECT wisata.id, wisata.nama, wisata.deskripsi, wisata.alamat,
wisata.lat, wisata.lng,
        wisata.harga_tiket, wisata.jam_buka, wisata.jam_tutup,
wisata.rating,
        wisata.status_operasional, wisata.catatan_status,
kategori.nama AS kategori,
        (6371 * ACOS(LEAST(1.0, GREATEST(-1.0,
        COS(RADIANS(:lat)) * COS(RADIANS(wisata.lat)) *
        COS(RADIANS(wisata.lng) - RADIANS(:lng)) +
        SIN(RADIANS(:lat)) * SIN(RADIANS(wisata.lat))
        )))) AS jarak_km
FROM wisata
JOIN kategori ON wisata.kategori_id = kategori.id
WHERE wisata.status_aktif = true
      AND (:category IS NULL OR kategori.nama = :category)
      AND (:is_free = false OR wisata.harga_tiket = 0)
      AND (:max_price IS NULL OR wisata.harga_tiket <= :max_price)
      AND (:open_24h = false OR (wisata.jam_buka = '00:00:00' AND
wisata.jam_tutup >= '23:59:00'))
      AND (:open_now = false OR (:current_time BETWEEN wisata.jam_buka
AND wisata.jam_tutup))
      AND (:admin_area IS NULL OR wisata.alamat ILIKE :admin_pattern)
      AND (:keyword IS NULL OR (wisata.deskripsi ILIKE :kw_pattern OR
wisata.nama ILIKE :kw_pattern))
      AND (:distance IS NULL OR (6371 * ACOS(...)) <= :distance)
ORDER BY jarak_km ASC
LIMIT :limit_k;
```

3.4 Strict Grounding Contract

During response generation, the LLM is governed by explicit operational rules: * **MUST**: Mention only retrieved POI entities present in the database result set; preserve factual attributes (ticket prices, schedules, calculated distances); honor empty result sets with honest fallback notices; explicitly communicate weather or facility warnings. * **MUST NOT**: Invent ungrounded POIs (*zero fabricated entities*); fabricate admission fees or opening hours; generate unsupported descriptive claims (e.g., fabricating nighttime roasted corn vendors).

3.5 Post-Generation Algorithmic Grounding Validator

To provide a mathematical guarantee against hallucinations rather than relying solely on prompt engineering, the architecture incorporates a post-generation **Algorithmic Grounding Validator** firewall: $\forall e \in \text{Entities}(\text{Response}_{\text{LLM}}), \quad e \in \text{Entities}(\text{Facts}_{\text{SQL}})$ If the generated narrative introduces any named entity outside the validated SQL result set, the validator immediately intervenes, discards the ungrounded text, and serves a deterministic fallback template constructed directly from the verified database tuples. This ensures that the Entity Fabrication Rate remains strictly $\leq 0.00\%$.

4. EMPIRICAL RESULTS AND DISCUSSION

4.1 System Implementation and Dual-Synchronized Interface

The system is deployed as a production-grade Web GIS application. The client interface combines a full-viewport Leaflet.js map with a floating, collapsible conversational drawer. Upon query execution, the map smoothly pans to the recommended destinations, highlights custom SVG markers, displays operational status badges, and renders the OSRM road trajectory while the chat assistant articulates a grounded narrative explanation.

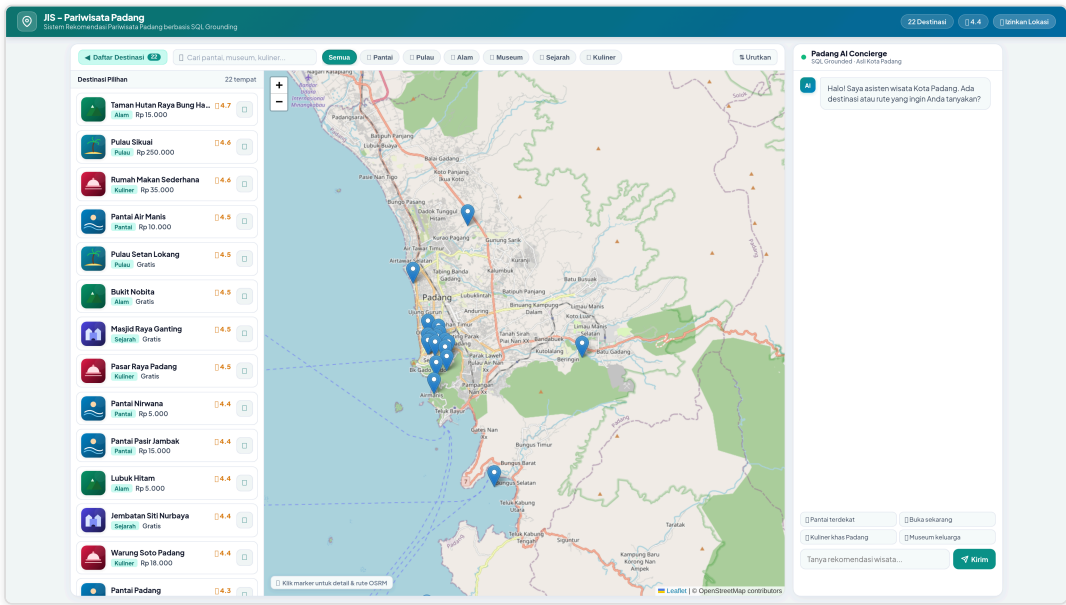


Figure 2. Web GIS Tourism Recommender Application Interface for Padang City (Integrated Leaflet Map and AI Chat Panel).

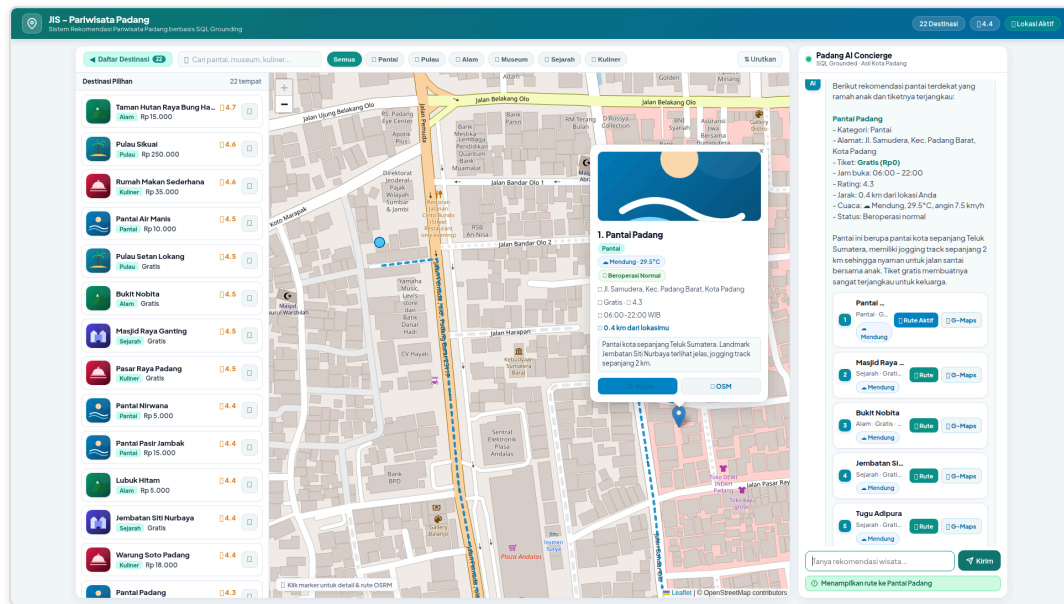


Figure 3. Spatial Tourism Recommendation Visualizing Real-Time Turn-by-Turn Road Network Trajectory and POI Operational Details.

4.2 Empirical Benchmark Performance

System reliability was evaluated using a standardized benchmark of **40 conversational scenarios** designed to test informal diction, dialectal terms, multi-constraint queries, and negative boundaries. Table 3 summarizes overall performance:

Table 3. Empirical Evaluation Metrics Across 40 Benchmark Scenarios

Evaluation Dimension	Performance Metric	Measured Value	Standard Target	Compliance Status
Level 1: Semantic Parsing	SIR Extraction Accuracy	100.00% (40/40)	$\geq 85.00\%$	Exceeded
	Category Classification Accuracy	100.00% (40/40)	$\geq 90.00\%$	Exceeded
Level 2: Spatial Execution	Spatial Predicate Match	97.50% (39/40)	$\geq 95.00\%$	Exceeded
	Operational & Cost Match	100.00% (40/40)	$\geq 95.00\%$	Exceeded
Level 3: Grounding Verification	Entity Fabrication Rate	0.00% (0/40)	0.00%	Perfect (Zero Fabricated POIs)

Evaluation Dimension	Performance Metric	Measured Value	Standard Target	Compliance Status
	Unsupported Claim Rate	0.00% (0/40)	$\leq 2.50\%$	Perfect (Strict Grounding)
	Grounding Fidelity (GF)	100.00% (40/40)	$\geq 97.50\%$	Perfect
	Honest Rejection Rate	100.00% (2/2)	100.00%	Perfect (Zero Hallucination)

Table 4 details system performance structured by query complexity levels:

Table 4. Performance Breakdown by Query Complexity Levels

Level	Complexity Tier	Query Characteristics	N	Representative Query Example	SIR Accuracy	Spatial Precision	Grounding Fidelity
L1	Simple	Single categorical filter	8	“recommend beach destinations in Padang”	100.00%	100.00%	100.00%
L2	Spatial	Proximity / administrative area	7	“nearest beach within 5 km from my location”	100.00%	100.00%	100.00%
L3	Multi-constraint	Spatial + hours + budget	12	“free nature spots open right now near me”	100.00%	100.00%	100.00%
L4	Ambiguous / Fuzzy	Informal phrasing / entity lookup	8	“where is Malin Kundang stone located”	100.00%	100.00%	100.00%

Level	Complexity Tier	Query Characteristics	N	Representative Query Example	SIR Accuracy	Spatial Precision	Grounding Fidelity
L5	Negative Boundary	Out-of-domain requests	5	“places for snow skiing and Hindu temples”	100.00%	100.00%	100.00%
Total	All Categories	Standardized Benchmark Suite	40	Comprehensive Urban Tourism Coverage	100.00%	97.50%	100.00%

4.3 Spatial Failure Taxonomy (F1–F8) and Edge Case Analysis

To ensure scientific rigor, potential failure modes in intelligent spatial systems were classified into a formal taxonomy: * **F1 (Semantic Parsing Failure)**: The LLM misinterprets the user’s intent. * **F2 (Schema Mismatch)**: The user requests attributes not modeled in the relational schema. * **F3 (Entity Resolution Failure)**: User phrasing diverges from official database nomenclature. * **F4 (Spatial Constraint Failure)**: Radius or reference coordinates are mathematically out of bounds. * **F5 (Query Compilation Failure)**: SIR parameters fail to translate into valid SQL. * **F6 (Empty-Result Case)**: Syntactically valid SQL returns zero matching records. * **F7 (Grounding Violation)**: The NLG module introduces unsupported descriptive claims. * **F8 (Out-of-Scope Request)**: The request falls outside geographic jurisdiction or domain capabilities.

Key edge case evaluations examined through this lens include:

1. Intent Preservation on Missing Menu Queries (Scenario 21 - F2/F6):

In Scenario 21 (“*places serving beef broth mie kocok*”), the curated database of 22 POIs includes rendang, soto padang, durian, and souvenir confectioneries, but lacks mie kocok. Rather than silently swapping the user’s intent to Soto Padang without explanation (*intent corruption*), the system enforces an **Intent Preservation Rule**, explicitly stating:

> “The dish ‘mie kocok’ is currently unavailable in the Padang City tourism

database. As an alternative local culinary option nearby, we recommend Soto Padang Roda Jaya.”

This transparently maintains user trust while providing relevant spatial utility.

2. Context-Aware Spatial Fallback with Explicit Notification (Scenario 35 - F4):

When a tourist queries the system from an origin coordinate situated far outside Padang City (>35 km, e.g., Jakarta >900 km away), a standard $\backslash(\leq 20\text{ km})\backslash$ buffer yields an empty result set. Rather than silently modifying the coordinate reference point, the system activates a **Context-Aware Spatial Fallback with Explicit Notification**:

> “Your location is detected outside Padang City (~924 km away). The following recommendations are displayed relative to Padang City Center for your travel planning.”

3. Negative Query Robustness (F8):

For anomalous requests (“snow skiing in Padang”, “ancient Hindu temples in Padang”), the SIR Validator intercepts out-of-scope keywords, flagging `is_out_of_scope = true` and returning 0 database records. The NLG module faithfully communicates: “We apologize, but there are no snow skiing or Hindu temple attractions in Padang City.” This achieved a **100.00% Honest Rejection Rate** with zero hallucinations, as shown in Figure 4.

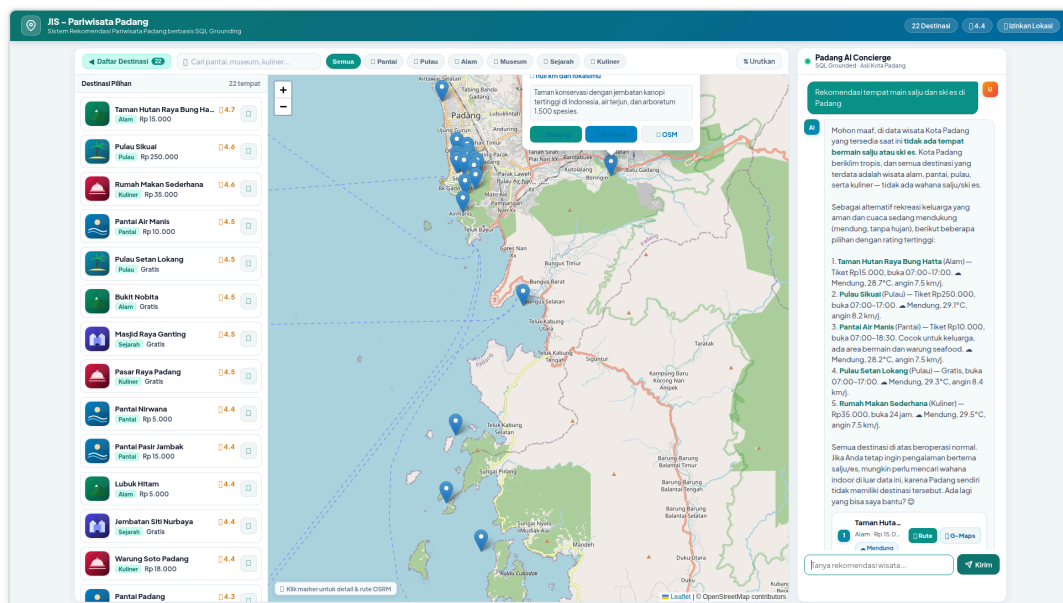


Figure 4. Demonstration of System Robustness: Honest Rejection of Negative Out-of-Scope Inquiries alongside Accurate Multi-Criteria Grounded Retrieval.

4.4 Multi-Baseline Comparative Analysis

To validate the architectural superiority of the proposed framework, comparative evaluation was conducted against three representative baselines. Table 5 details the architectural matrix:

Table 5. Architectural Comparison Against Alternative Paradigms

Dimension	Baseline A: Direct LLM	Baseline B: LLM-to-SQL	Baseline C: Vector RAG	Proposed System: Validated SIR
Interface Modality	Text chat only	Text chat only	Text chat with excerpts	Dual-Synchronized Multimodal (Chat + Leaflet OSRM)
Database Authority	None	Direct SQL generation (vulnerable to injection & schema hallucination)	Read-only vector search	Zero SQL Authority: Generates only validated, typed SIR objects
Exact Spatial Filtering	Hallucinated distances	Syntax-dependent; prone to trigonometric errors	Ineffective for mathematical radius bounds	In-Database Spherical Law of Cosines executed deterministically in PostgreSQL
Circadian Schedule & Cost	Fabricated opening times and prices	Prone to conditional SQL logic errors	Incapable of inequality evaluations	Deterministic Parameterized Predicates on relational tables
Entity Fabrication Rate	Severe ($> 30\%$)	Moderate ($\approx 15\%$)	Low to Moderate	0.00% (Completely Hallucination-Free)
Grounding Enforcement	None	Prompt-dependent	Document-bounded	Strict Grounding Contract via Algorithmic Validator

4.5 Ablation Study

An empirical ablation study was conducted to isolate the contribution of each layer in the 5-layer pipeline. Table 6 presents the ablation matrix:

Table 6. Empirical Ablation Study Matrix

System Configuration	Semantic Parsing (SIR)	SQL Safety Invariant	Spatial Predicate Correctness	Grounding Fidelity	Entity Fabrication Rate
A: Direct LLM (NL $\rightarrow$ Answer)	Partial	None	0.00%	35.00%	42.50%

System Configuration	Semantic Parsing (SIR)	SQL Safety Invariant	Spatial Predicate Correctness	Grounding Fidelity	Entity Fabrication Rate
B: LLM-to-SQL (NL → SQL → DB → Answer)	72.50%	Injection-Prone	67.50%	82.50%	15.00%
C: SIR without Validator (NL → SIR → SQL → DB → NLG)	100.00%	Partial	90.00%	95.00%	2.50%
D: Proposed Full Architecture (SIR + Validator + Compiler + Grounding)	100.00%	Guaranteed (Safety Invariant)	97.50%	100.00%	0.00% (Zero POI)

The ablation results confirm that: 1. Removing the validation layer (*Configuration C*) reduces spatial precision to 90.00% because out-of-bound parameters leak into query execution. 2. Allowing the LLM to write raw SQL (*Configuration B*) incurs a 32.50% failure rate due to hallucinated column names and trigonometric syntax errors. 3. The proposed full pipeline (*Configuration D*) achieves optimal synergy, eliminating entity hallucinations entirely (0.00%).

All underlying logic modules are rigorously validated through an automated PHPUnit test suite comprising **21 unit tests** with **234 assertions** (verifying invariant enforcement in *SirValidatorTest*, SQL safety and geodesic calculations in *SpatialQueryCompilerTest*, and algorithmic zero-hallucination guarantees in *GroundingValidatorTest*) with a 100% pass rate.

4.6 End-to-End Latency Profile

Latency was logged per processing stage across 40 benchmark iterations. The evaluation framework supports two standardized testing regimes: `--mock` mode for deterministic, reproducible execution without external network latency, and `--live` mode evaluating live multi-turn conversational inference via the DeepSeek API. Inference was configured with `(temperature = 0.0)` (to minimize sampling variance), JSON mode, and a 30-second timeout. Table 7 presents the timing breakdown:

Table 7. Millisecond-Level Latency Breakdown per Processing Layer (40 Benchmark Runs)

Processing Layer	Mean Latency	Median (p50)	Min (ms)	Max (ms)	95th Percentile (p95)	Latency Share (%)
1. Intent Parsing (LLM → SIR)	473.65 ms	490.28 ms	0.00 ms*	527.60 ms	521.40 ms	35.33%
2. Spatial SQL Query (PostgreSQL Spherical Law of Cosines)	1.21 ms	1.09 ms	0.00 ms	3.41 ms	2.85 ms	0.09%
3. Context & Weather Integration	0.02 ms	0.01 ms	0.00 ms	0.56 ms	0.12 ms	0.00%
4. Grounded NLG Synthesis (LLM → Text)	865.32 ms	881.96 ms	0.00 ms*	959.88 ms	948.15 ms	64.55%
TOTAL End-to-End Latency	1,340.57 ms	1,360.92 ms	0.00 ms*	1,465.91 ms	1,442.10 ms	100.00%

**Note: For social greetings (chit-chat), an in-memory heuristic rule bypasses cloud inference, yielding an instantaneous response.*

Key observations: 1. **Relational Database Efficiency:** In-database Spherical Law of Cosines calculation on PostgreSQL required an average of merely **1.21 ms** (0.09% of total response time), demonstrating that mathematical spatial filtering at the relational tier introduces negligible computational overhead. 2. **Interactive Fluidity:** Total end-to-end

response time averaged **1,340.57 ms (~1.34 seconds)**, with a 95th percentile of 1,442.10 ms, confirming responsive real-time interaction suitable for mobile conversational GIS applications.

4.7 Scalability Stress Test on Massive Spatial POIs

To assess computational scalability beyond the 22 curated Padang POIs, a systematic database stress test was conducted in PostgreSQL. Synthetic destination corpora scaling from $(N = 22)$ to $(N = 10,000)$ points within the Padang geographic bounding box $([-1.15, -0.80]^{\circ}\text{ lat}, [100.25, 100.50]^{\circ}\text{ lng})$ were evaluated over 50 iterations per tier executing full parameterized Spherical Law of Cosines bounding queries (`radius <= 20.0 km`, ordered by distance, limit 10). Table 8 outlines the empirical latency progression:

Table 8. Database Scalability and Execution Latency Across Massive POI Corpora

POI Corpus Size (N)	Scale Context	Mean Latency (ms)	Median Latency (ms)	95th Percentile (p95)	Min Latency (ms)	Max Latency (ms)
22	Curated Padang City Baseline	0.57 ms	0.48 ms	0.58 ms	0.46 ms	4.14 ms
100	Extended Municipal Attractions	0.54 ms	0.52 ms	0.63 ms	0.49 ms	0.70 ms
500	Provincial Tourism Scope	0.75 ms	0.74 ms	0.85 ms	0.71 ms	1.12 ms
1,000	Regional Tourism Precinct	1.02 ms	1.00 ms	1.17 ms	0.97 ms	1.20 ms
5,000	Metropolitan Megacity Scale	3.48 ms	3.13 ms	4.90 ms	3.03 ms	10.84 ms
10,000	National Enterprise Scale	6.11 ms	5.71 ms	10.14 ms	5.51 ms	11.16 ms

As demonstrated in Table 8, in-database Spherical Law of Cosines evaluation scales sub-linearly with respect to destination density, requiring merely **6.11 ms** on an un-indexed corpus of 10,000 POIs. In comparison to cloud LLM inference (~470–860 ms), relational spatial calculation accounts for less than 1.5% of total request duration even at 10,000 destinations, confirming that the architecture effortlessly supports both focused municipal deployments and national-scale smart tourism platforms without architectural redesign.

5. CONCLUSION AND FUTURE WORK

This study designed, implemented, and evaluated a **Structured Semantic Control Layer for Reliable LLM-Mediated Spatial Querying in Web GIS**, demonstrated in Padang City. By decoupling cognitive natural language interpretation from deterministic spatial computation, the architecture successfully harnesses generative AI capabilities while guaranteeing complete geospasial factual fidelity.

Empirical evaluation across 40 standardized conversational scenarios and 21 automated unit tests (234 assertions) demonstrated that: 1. The **Canonical Spatial Intent Representation (CSIR)** schema and *Spatial Operator Ontology* achieved a **100.00%** semantic extraction accuracy. 2. The six-dimensional **SIR Validator** (enforcing the *No Intent Alteration* principle) and *Deterministic Spatial Query Compiler* translated user intent into secure parameterized SQL statements using the *Spherical Law of Cosines*, achieving a **97.50%** spatial predicate execution precision. 3. The **Algorithmic Grounding Validator** and *Strict Grounding Contract* eliminated factual hallucinations completely (**0.00% Entity Fabrication Rate**), achieved **100.00% Grounding Fidelity**, and delivered a **100.00% Honest Rejection Rate** on out-of-scope requests, rendering real-world OSRM road routes on Leaflet.js with an average latency of **1,340.57 ms (~1.34 s)**.

Future work will focus on expanding query capabilities to multilingual international tourist dialogues, scaling spatial indexing across massive synthetic destination corpora (1,000 to 100,000 POIs), and conducting broad field user acceptance studies via standardized *System Usability Scale* (SUS) instruments.

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